Survival Analysis of Loan Charge-offs Using Cox Regression

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Introduction

How can a bank choose which loans to accept and reject while balancing loss due to charge offs and opportunity costs?

Goal: Model the probability of a loan charging off at various times throughout the life of the loan to assist in deciding if a borrower is worth lending to.

Method: Cox regression. It's a model that's used in survival analysis to determine how different factors impact survival time

The Data

State Bank of Southern Utah's loan charge-off data from 1976 to 2026

What is a loan charge-off?
When a creditor deems a loan is uncollectable, usually 3 to 6 months after non-payment.

Limitations: State Bank of Southern Utah doesn't have a dataset with all loans from this period, with charge off dates included.

Remedial Measures

Because State Bank of Southern Utah doesn't have a data set with all loans, but just the loans that charged off, we need to alter the dataset to get it to work with Cox regression.

- Censor future loans to make use of that feature of Cox regression
- Remove loans with an original maturity beyond 2026 to avoid survivorship bias
- Change interpretations to only include loans that will eventually charge off.

The Data: Attributes

- **Application Code (categorical):** The application codes indicate the type of loan on a broad perspective, Commercial Loans (CC), Installment or traditional Consumer Loans (CI), or Real Estate (CR)
- **Product (categorical):** the type of loan issued.
- **Interest Rate (continuous):** the effective annual rate associated with the loan.
- **Original Note Date (discrete):** The date the loan was issued.
- **Original Maturity Date (discrete):** The date the loan was intended to mature.
- **Current Maturity Date (Discrete):** The maturity date after loan modifications to assist payment.

The Data: Attributes

- **Charge Off Date (Discrete):**
  The date the loan was charged off.
- **Original Loan Amount (continuous):**
  The amount borrowed.
- **Total Chargeoff (continuous):**
  The amount charged off, or deemed uncollectible at the Charge Off Date.
- **LTD Recovery Amount (continuous):**
  The amount that was collected after the loan was charged off up until the data was retrieved.

Interest rate

The interest rate is one of the most important variables in our model. It contains information on:

- If the borrower has defaulted in the past
- Borrower's credit score
- Borrower's ability to make payments
- General economic conditions
- The overall risk of the borrower

Monthly Payment

We expect the monthly payment to be a good predictor too, since a higher monthly payment will be more difficult to repay. However this will vary by the borrower's:

- Income
- Other debts

Time to Maturity

The time to maturity will have a significant predictive effect in how long the loan survives for.

A 30 year mortgage will have more time to charge off than a 1 year auto loan.

Key variables

Predictor Variables

Interest Rate
Monthly Payment
Time to Maturity
Loan Amount
Application Code
Product

Response Variable

Time to Charge Off

What is Cox Regression

- Used in survival analysis to help us understand how different factors affect survival time.
- Survival analysis: how long it takes for an event to happen (chargeoff)
- We need to know an amount of time before the event happens (interval from start of loan to chargeoff)

What is Cox Regression

- Cox regression models hazard functions
- **Hazard function:** the instantaneous rate of death/failure given that the case has survived up to that point
- **Censored data:** cases where the event of interest (chargeoff) does not yet happen during our time of study. This data can still be useful. In the dataset, we must mark if a case is censored or uncensored data.

Math of Cox Regression

Time before event:

$$T \geq 0$$

Probability density function:

Frequency of event per period of time

$$f(t) = -\frac{dS(t)}{dt}$$

Survival function:

Probability a case survives longer than some time

$$S(t) = P(T > t), 0 < t < \infty$$

Hazard function:

The risk of the event occurring if it hasn't yet

$$h(t) = lim_{x \to \infty} \frac{P(t \leq T + \delta t | T \geq t)}{\delta t}$$

Math of Cox Regression

$$H(t) = \int_0^t h(u)\,du$$

Cumulative Hazard Function:
Cumulative risk up to a certain point

$$S(t) = e^{-H(t)}$$

Relationship between cumulative hazard and survival:
Higher hazards -> lower survival probability

$$\delta_i$$

Censoring indicator:
0 if i is uncensored, 1 if i is censored

$$t_i$$

Time of event or censoring

Math of Cox Regression

$$L = \Pi_{j had event} f(t_j) \Pi_{k censored} S(t_k) = \Pi_{i=1}^{N} h(t_i)^{\delta_i} S(t_i)$$

Likelihood function:

Set up so that cases for which we have the time of event are used to calculate the hazard rate at that time and survival to that time.

Cox Proportional Hazards Model

Single predictor $$h(t;x) = h_0(t)e^{\beta x}$$

Multiple predictors $$h(t;x_1,\ldots,x_p) = h_0(t)e^{\beta_1 x_1+\ldots+\beta_p x_p}$$

Interpretation: As x_k increases by one unit, the relative risk (hazard) is multiplied by exp(β_k), holding all else constant. (Covariates have an additive effect on the log hazard ratio

Covariate: x_k
Parameter: β_k

Math of Cox Regression

Beta values are estimated via partial likelihood

$$PL(\beta) = \Pi_{t_i:event\ at\ t_i} \frac{h_0(t)e^{\beta x_{(t_i)}}}{\sum_{j \epsilon R_{(t_i)}} h_0(t)e^{\beta x_j}}$$

$R_{(t_i)}$

Risk set: Cases that have never been censored nor had the event occur before time t_i

$x_{(t_i)}$

Value of covariate for the case for which the event occurred at time t_i

The probability it happened in that case instead of other potential cases

Math of Cox Regression

The baseline hazard can be canceled out, leaving:

$$PL(\beta) = \Pi_{t_i:\ event\ \ at\ \ t_i} \frac{e^{\beta x_{(t_i)}}}{\sum_{j:\ t_j \geq t_i} e^{\beta x_j}}$$

We treat this as a likelihood. The standard errors for beta estimates are calculated based on normal distribution.

Assumptions of Cox Regression

Ratio between overall hazard and baseline hazard function remains constant over time. Can be checked via:

- Test using χ^2 and Schoenfeld residuals

  H_O: Baseline hazard is proportional to the hazard function over time t.

  H_A: Baseline hazard is not proportional to the hazard function over time t
- Graph of scaled Schoenfeld residuals over time
  - Should be random scatter

What if Assumptions Aren't Met?

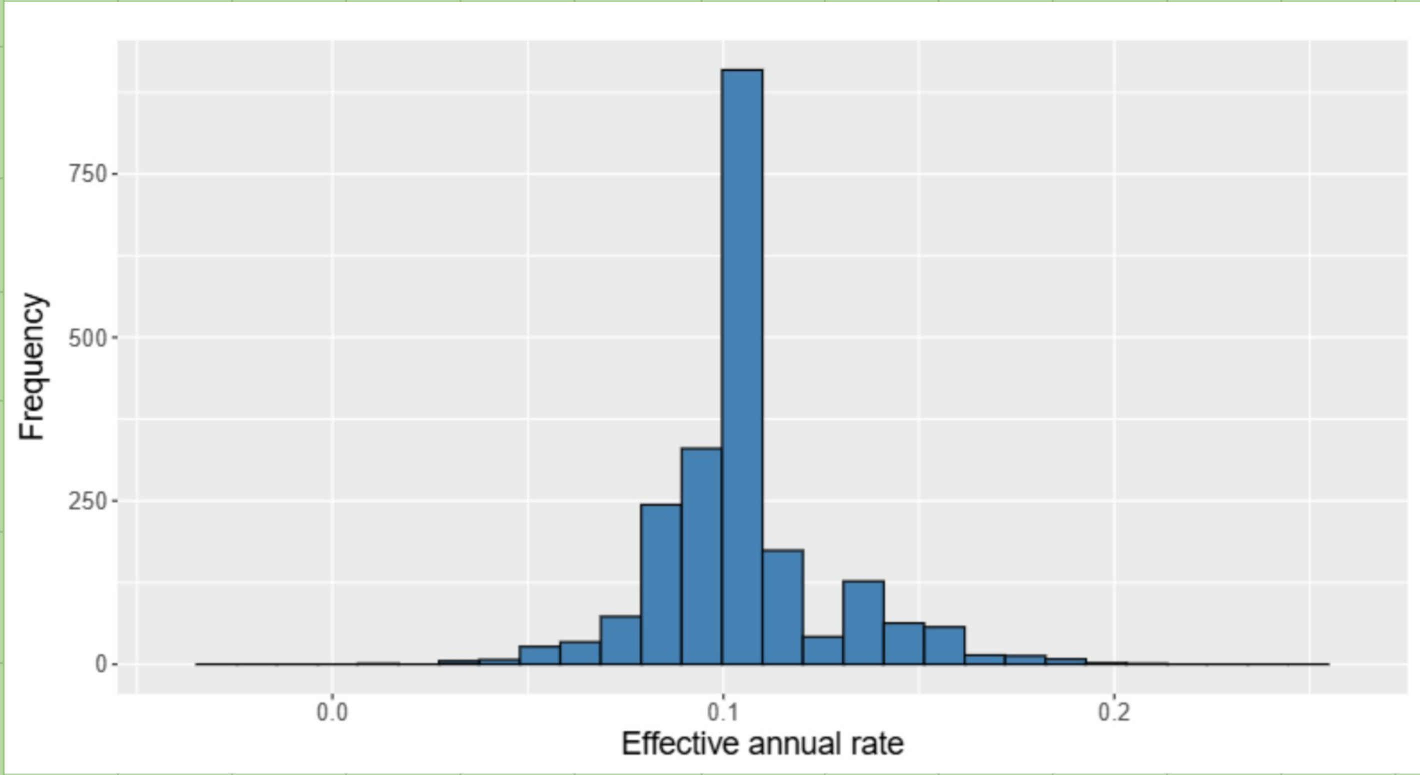
- Change the covariates
- Stratify by a categorical variable
  - Separate individuals based on some category
  - Trade-off: we can't compare groups to each other

Cleaning the Data

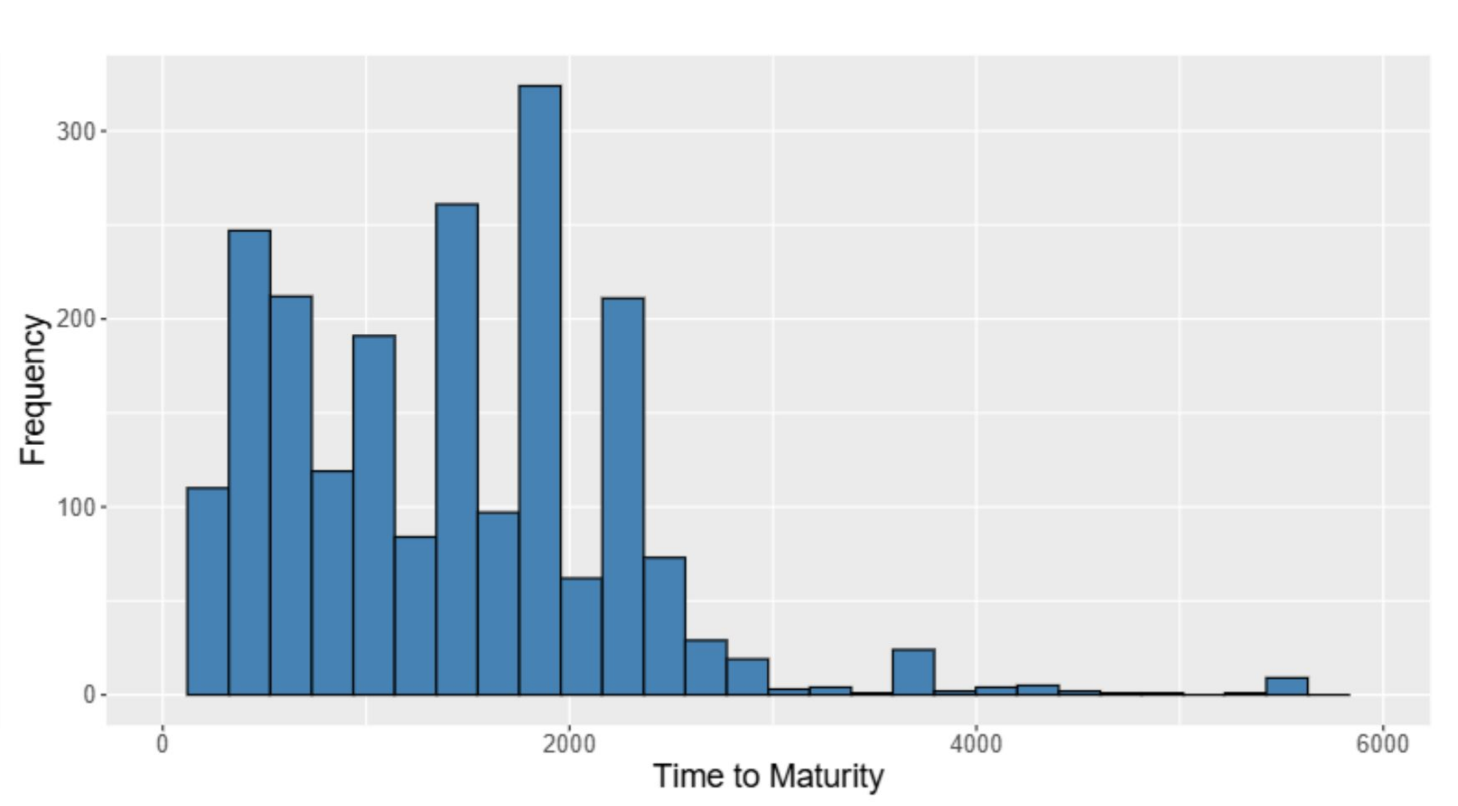
- Deleted “Chargeoff Surecaseh/Visa”, “Surecash”, “Interest Only Home Equity”, and “Home Equity 97” variables under the Product column
  - They do not have accurate Maturity Dates or have no end date.
- Deleted cases with zero interest rates
- Deleted loans with original maturity date after 3/5/2026 (the date we retrieved the data)

SureCash
Send Money Thru Phone

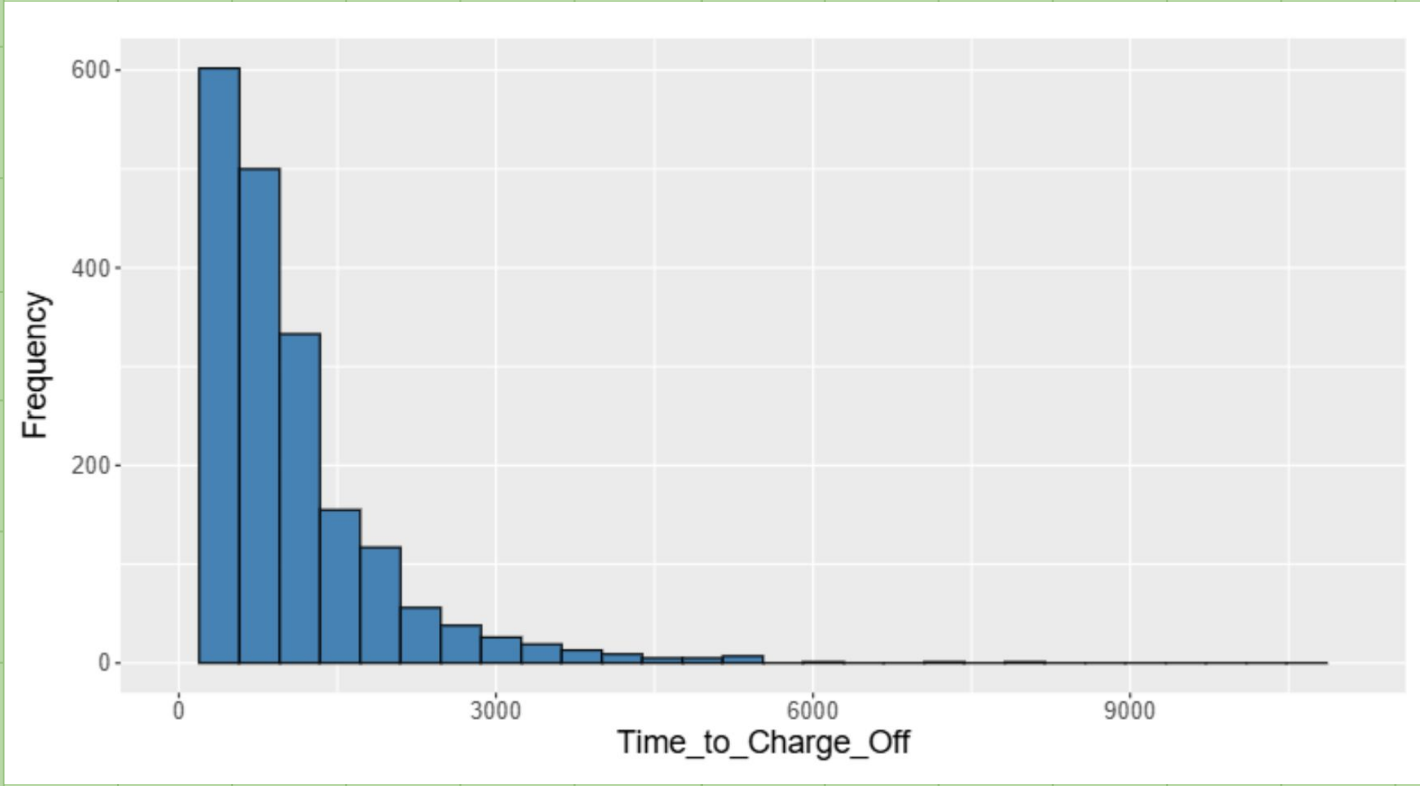
Histogram of EAR



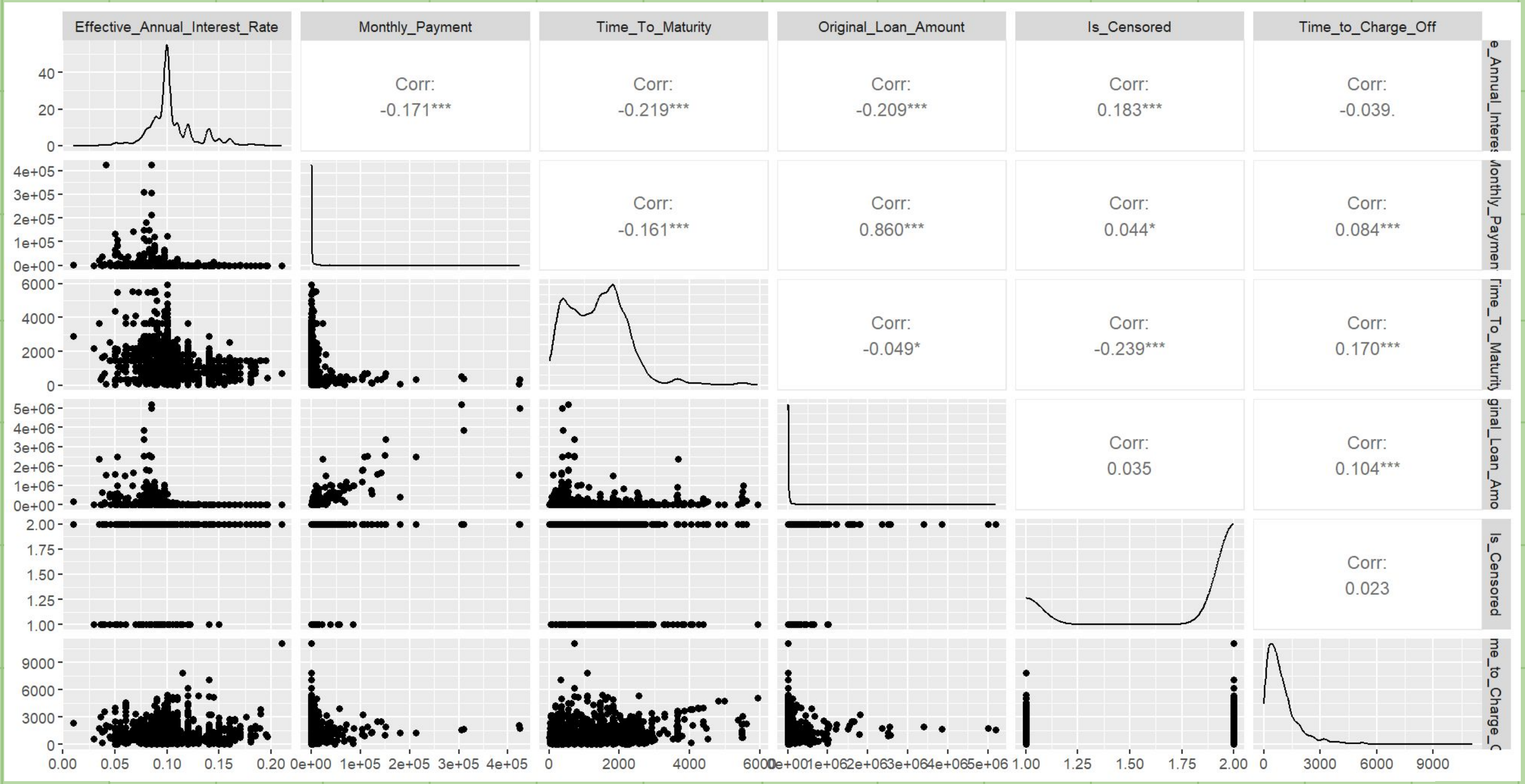
Histogram of Time to Maturity (days)



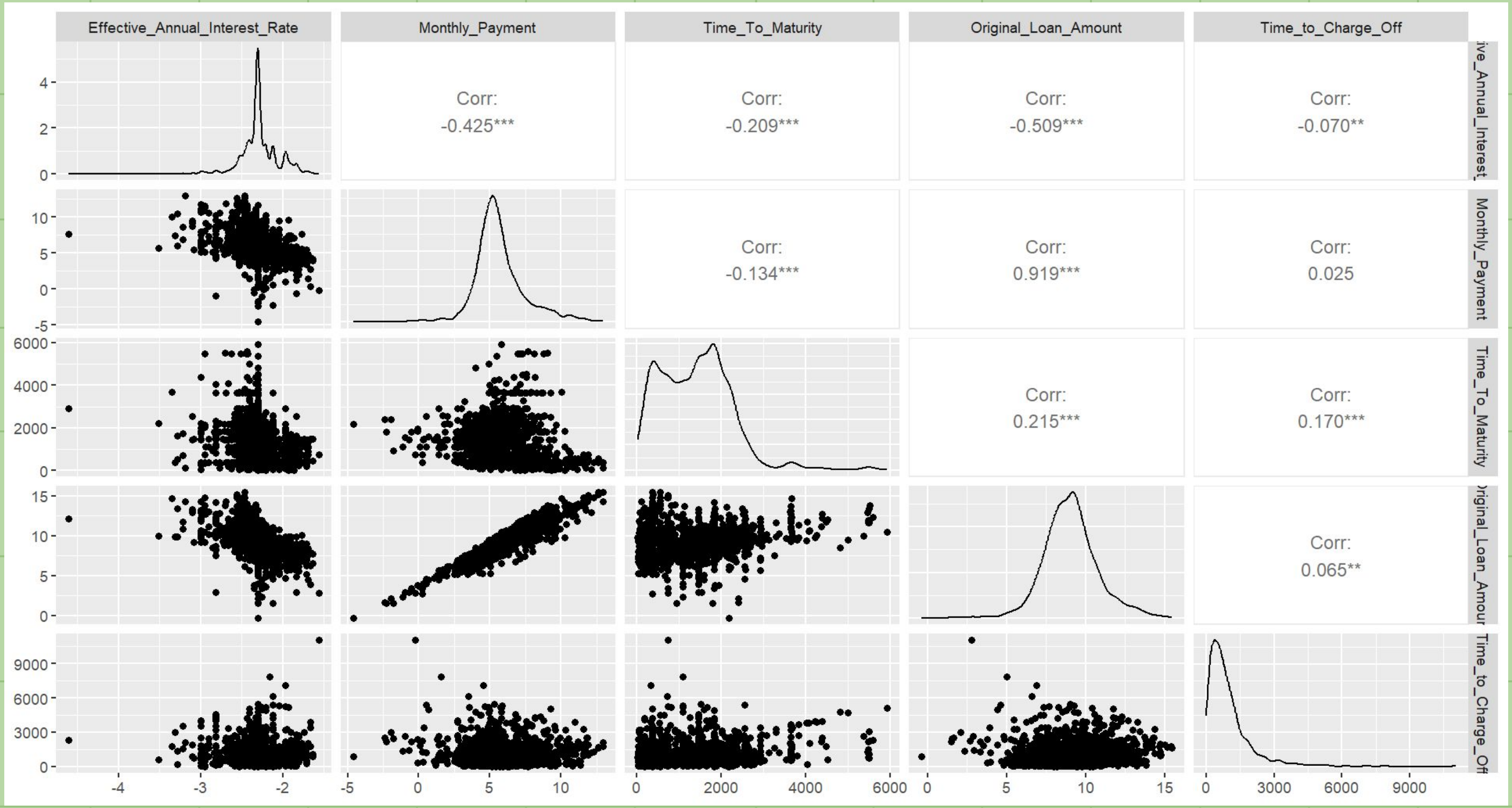
Histogram of Time to Charge Off (days)



Pairplot of Data



Pairplot with Logged Values



Fitting the Model

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	3.518e-01	1.422e+00	1.161e-01	3.030	0.00244
Monthly_Payment	-9.687e-02	9.077e-01	6.857e-02	-1.413	0.15773
Time_To_Maturity	-3.786e-04	9.996e-01	6.233e-05	-6.074	1.25e-09
Original_Loan_Amount	9.777e-02	1.103e+00	7.042e-02	1.388	0.16501
Application_CodeCI	1.316e-01	1.141e+00	7.188e-02	1.831	0.06716
Application_CodeCR	-1.319e-01	8.764e-01	1.131e-01	-1.166	0.24376

Likelihood ratio test=163.5 on 6 df, p=< 2.2e-16
n= 2131, number of events= 1688

Checking the Proportional Hazards Assumption

	chisq	df	p
Effective_Annual_Interest_Rate	14.3	1	0.00015
Monthly_Payment	11.3	1	0.00079
Time_To_Maturity	104.6	1	< 2e-16
Original_Loan_Amount	49.0	1	2.5e-12
Application_Code	74.5	2	< 2e-16
GLOBAL	199.1	6	< 2e-16

Stratifying the Model

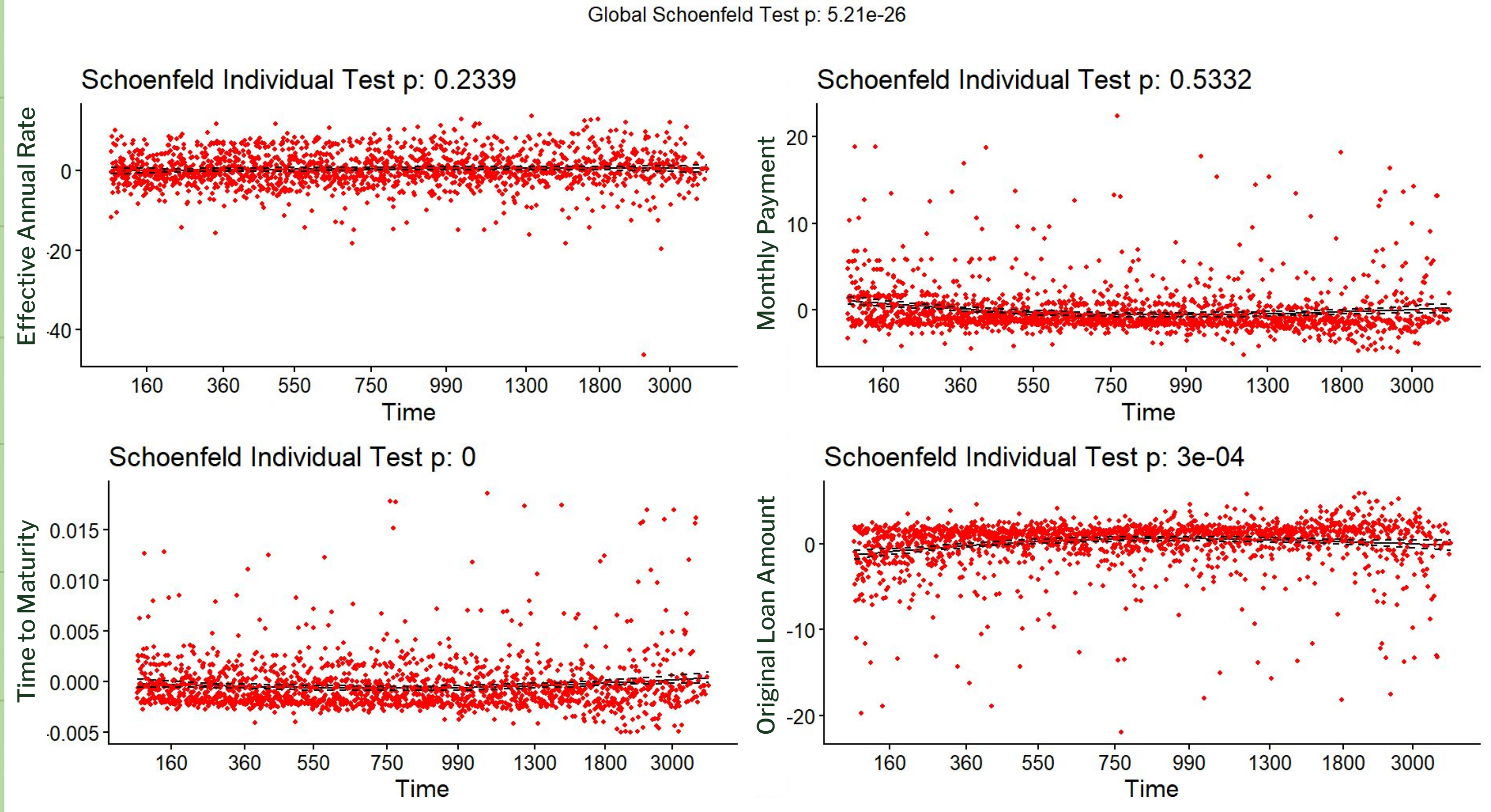
	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.3499141	1.4189457	0.1162197	3.011	0.00261
Monthly_Payment	-0.0937585	0.9105027	0.0684711	-1.369	0.17090
Time_To_Maturity	-0.0004059	0.9995941	0.0000625	-6.495	8.33e-11
Original_Loan_Amount	0.0984209	1.1034271	0.0704070	1.398	0.16215

Likelihood ratio test=156 on 4 df, p=< 2.2e-16
n= 2131, number of events= 1688

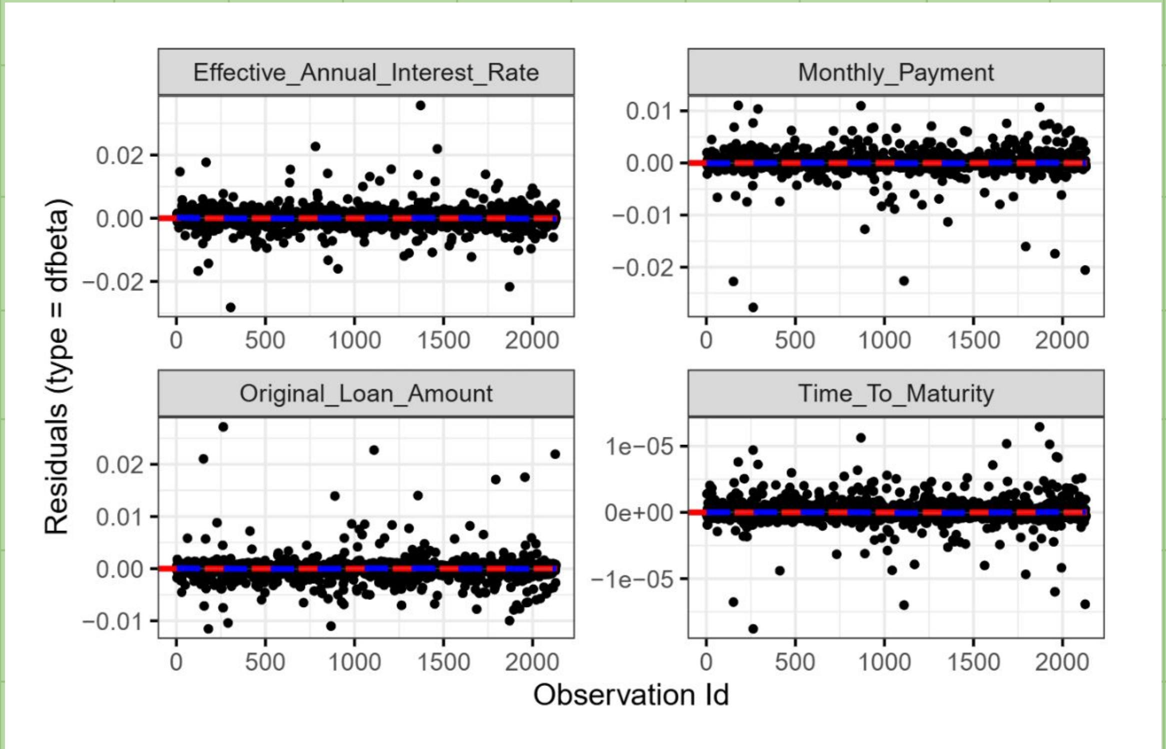
Proportional Hazards Assumption of Stratified Model

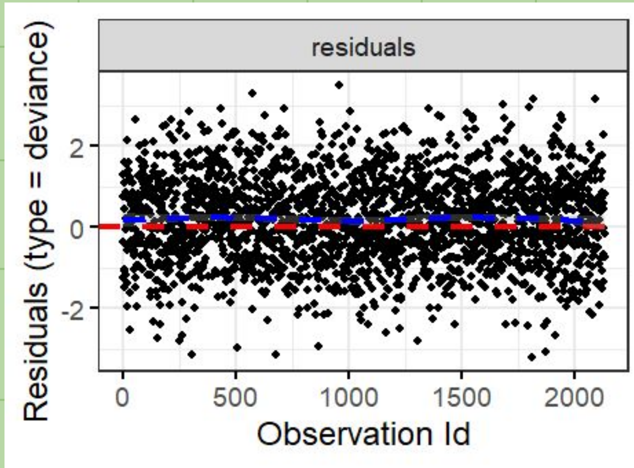
	chisq	df	p
Effective_Annual_Interest_Rate	1.417	1	0.2339
Monthly_Payment	0.388	1	0.5332
Time_To_Maturity	101.058	1	<2e-16
Original_Loan_Amount	13.055	1	0.0003
GLOBAL	124.731	4	<2e-16

Schoenfeld Residual Plot

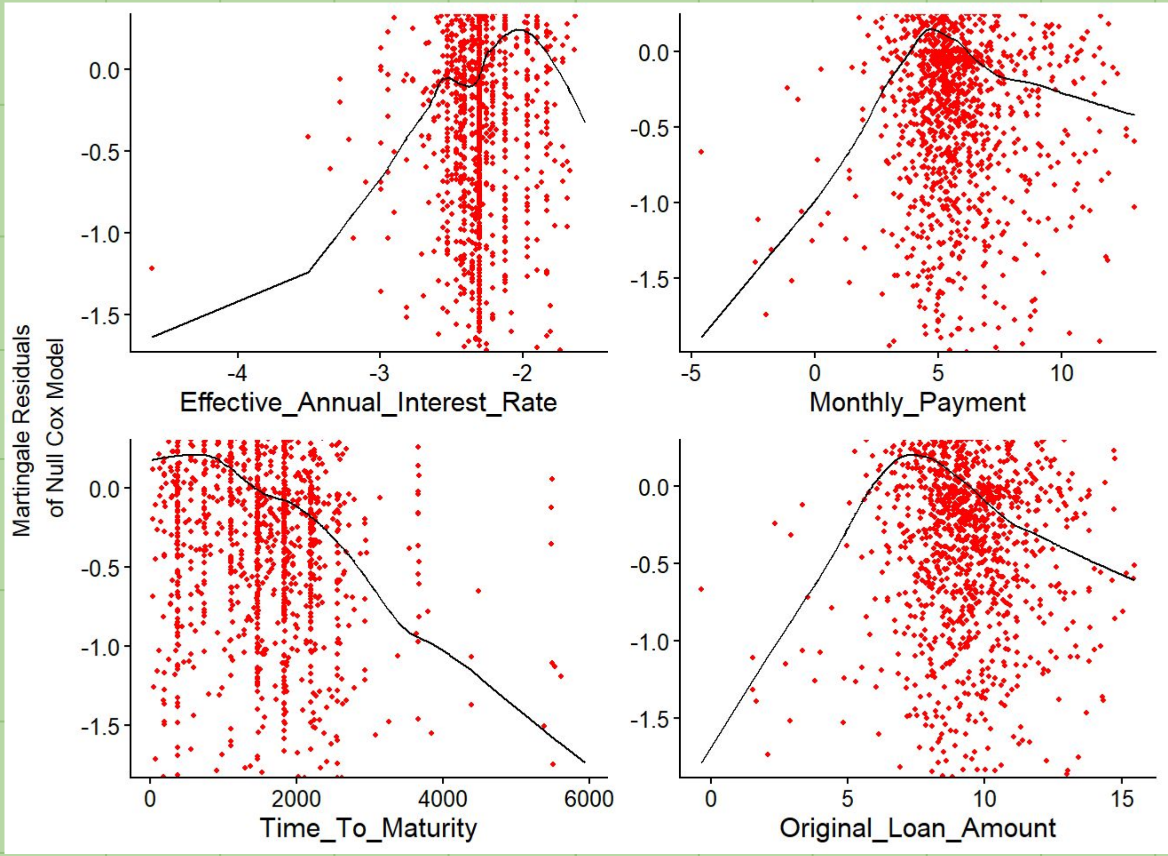


Influence Plot





Checking the Linearity Assumption



Checking for Multicollinearity

	GVIF	Df	GVIF^(1/(2*Df))
Effective_Annual_Interest_Rate	1.270015	1	1.126949
Monthly_Payment	27.521596	1	5.246103
Time_To_Maturity	4.090121	1	2.022405
Original_Loan_Amount	26.377343	1	5.135888
strata(Application_Code)	107.107496	0	Inf

Fitting the Model without Time_To_Maturity and Original_Loan_Amount

Model Summary

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.67313	1.96036	0.11452	5.878	4.16e-09
Monthly_Payment	0.05381	1.05528	0.01476	3.646	0.000267

Likelihood ratio test=38.12 on 2 df, p=5.289e-09
n= 2131, number of events= 1688

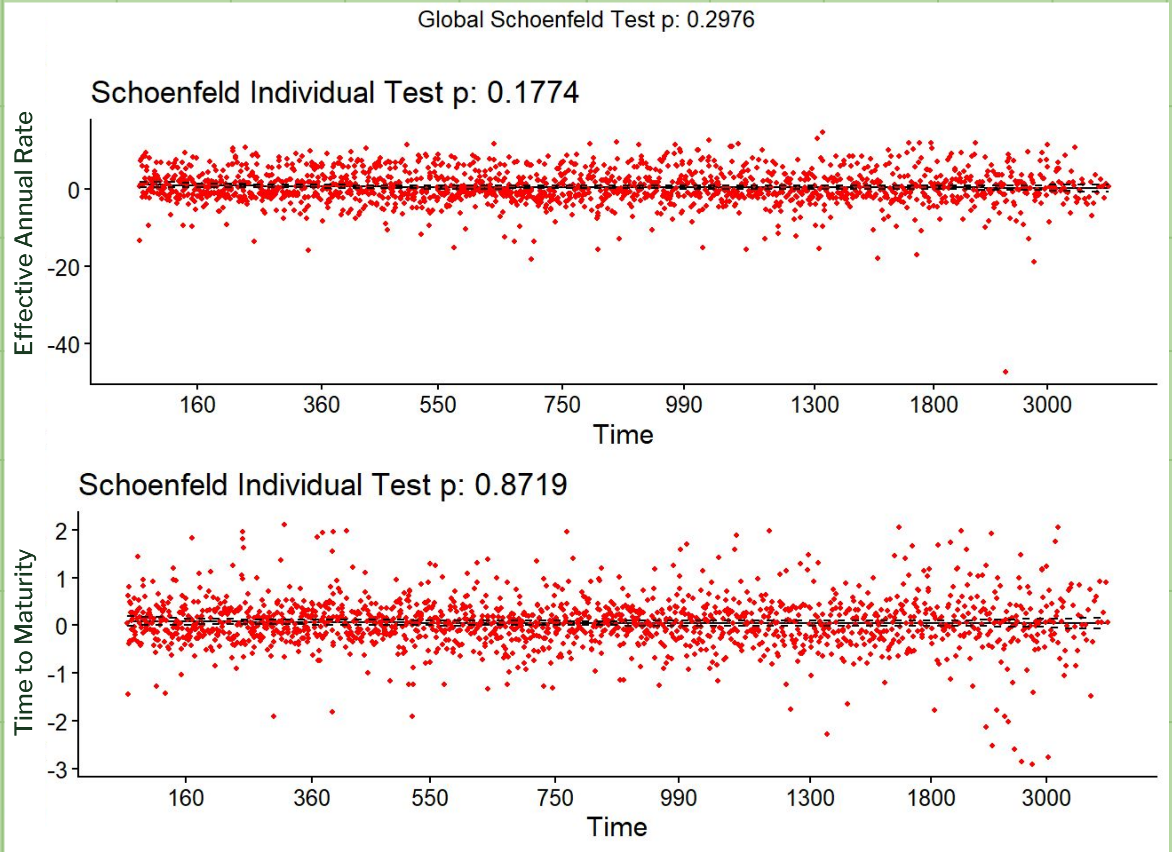
VIF values

	GVIF	Df	GVIF^(1/(2*Df))
Effective_Annual_Interest_Rate	1.176627	1	1.084724
Monthly_Payment	1.176627	1	1.084724
strata(Application_Code)	1.176627	0	Inf

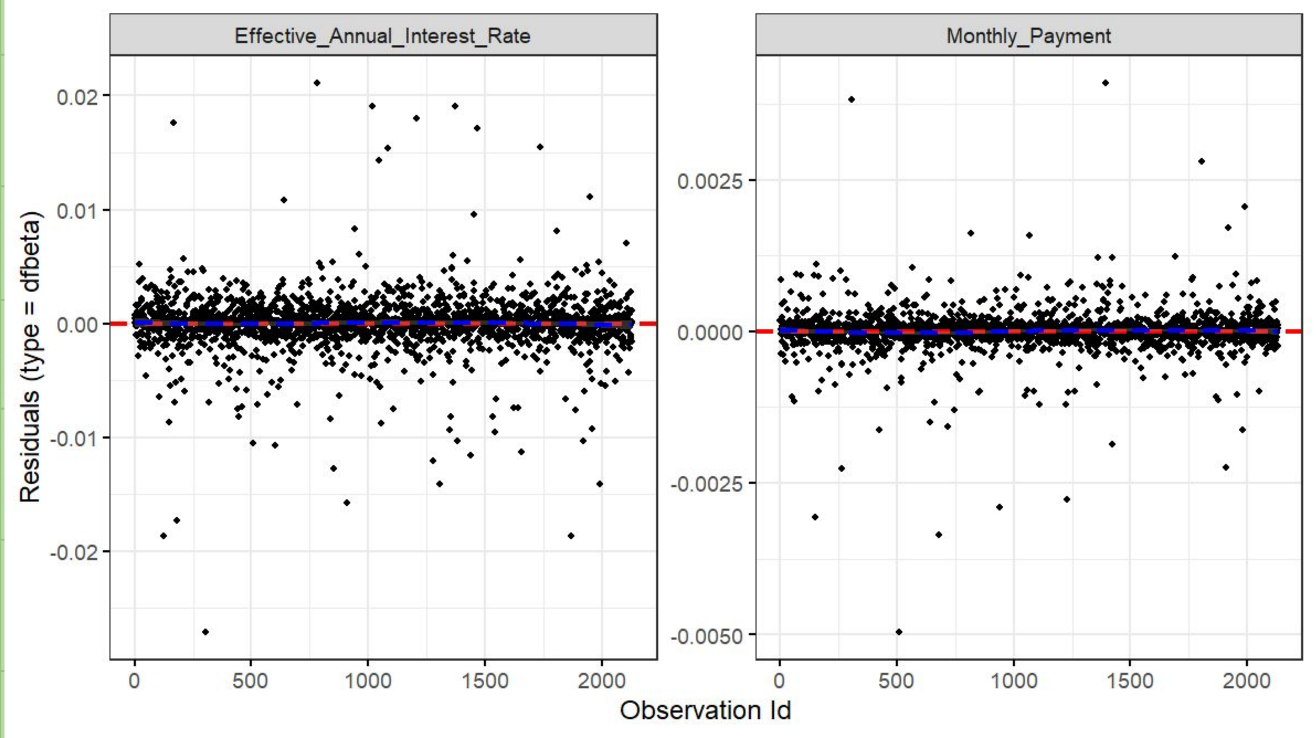
Checking Proportional Hazards Assumption

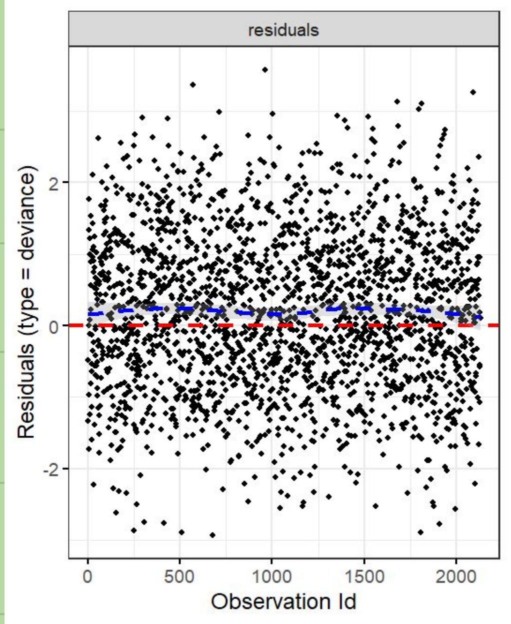
	chisq	df	p
Effective_Annual_Interest_Rate	1.819	1	0.18
Monthly_Payment	0.026	1	0.87
GLOBAL	2.424	2	0.30

Schoenfeld Residual Plot

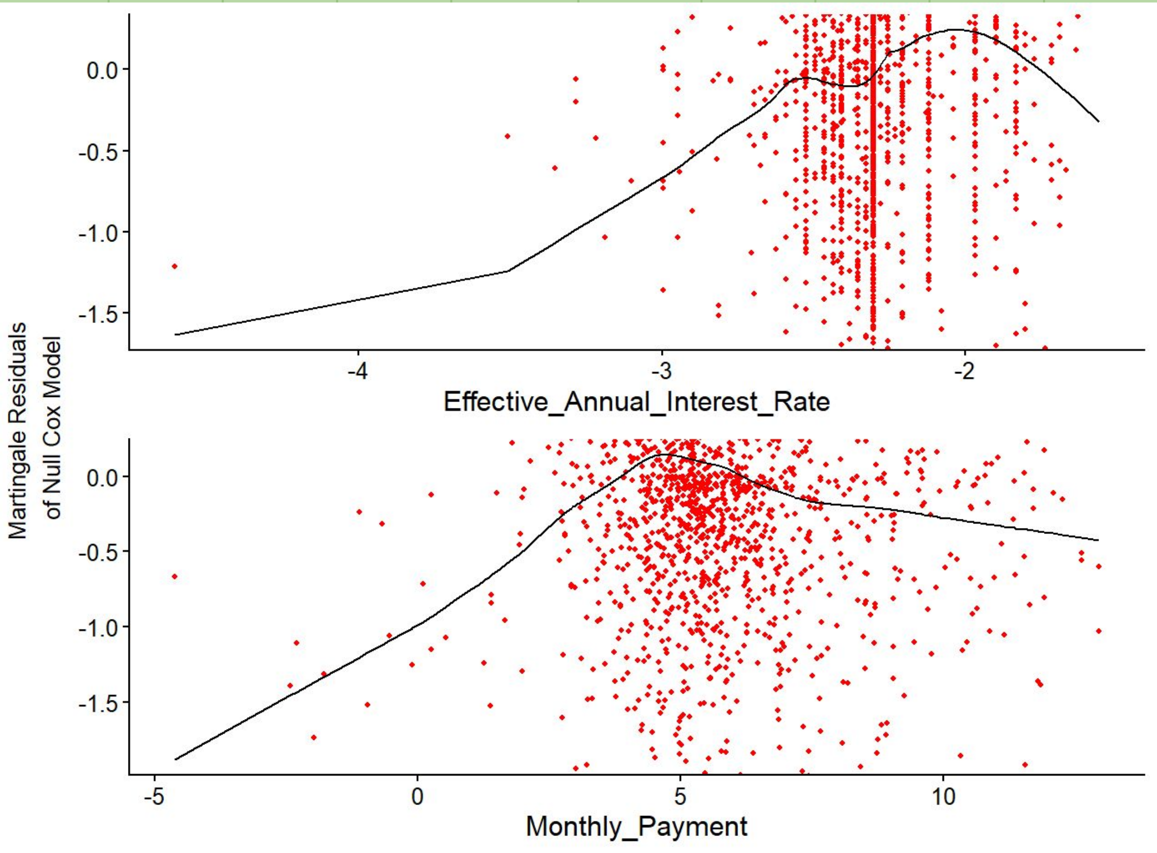


Influence Plot

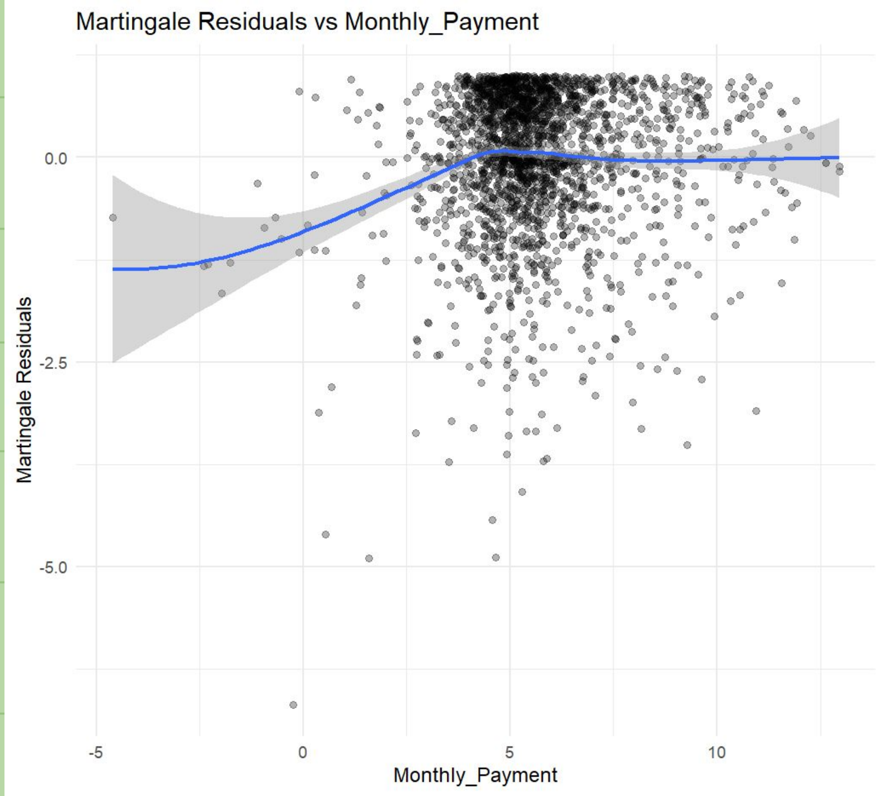


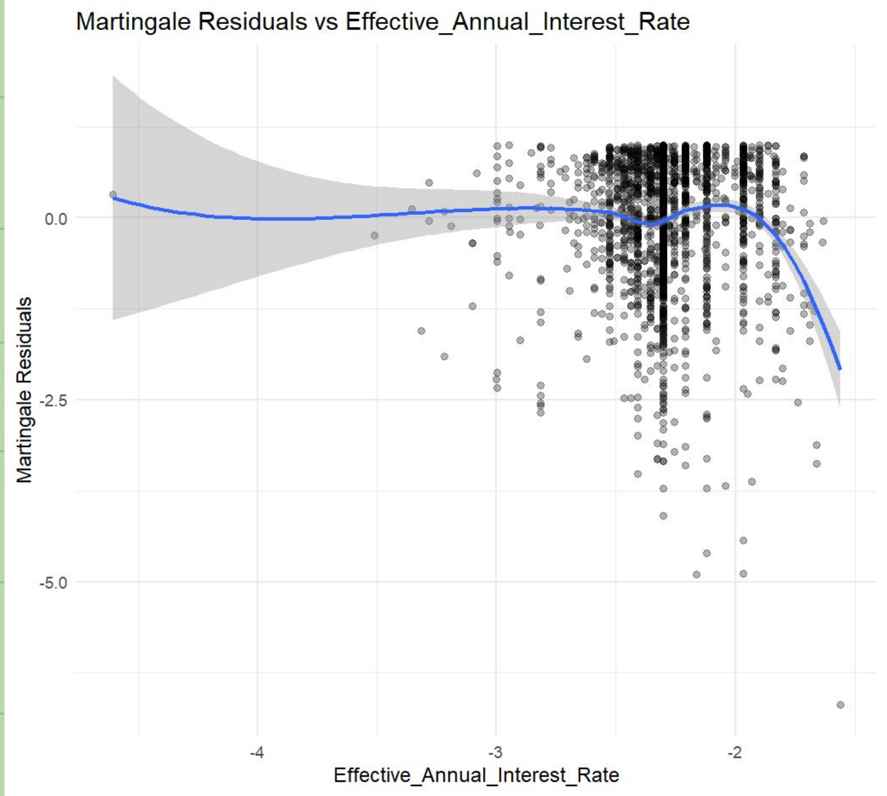


Checking Linearity Assumption



Checking Linearity Assumption via Martingale Residuals





Stratified Model

Earlier we stratified our model. This only works if the coefficients are similar. Let’s check the coefficients.

Stratified Models

Commercial Loans

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.67533	1.96469	0.32173	2.099	0.03581
Monthly_Payment	0.12759	1.13609	0.04158	3.068	0.00215

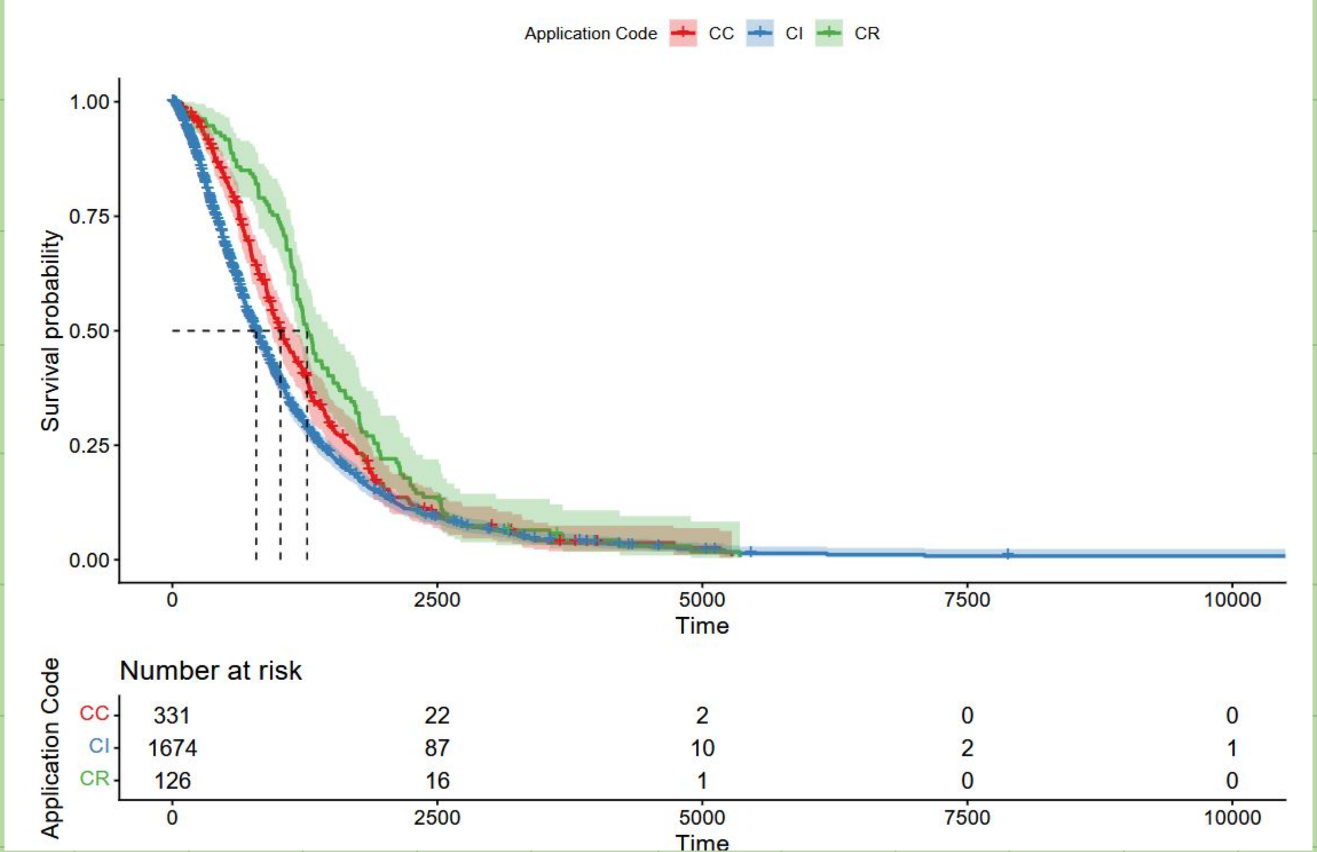
Real Estate

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.57815	1.78274	0.45955	1.258	0.208
Monthly_Payment	0.11777	1.12499	0.04795	2.456	0.014

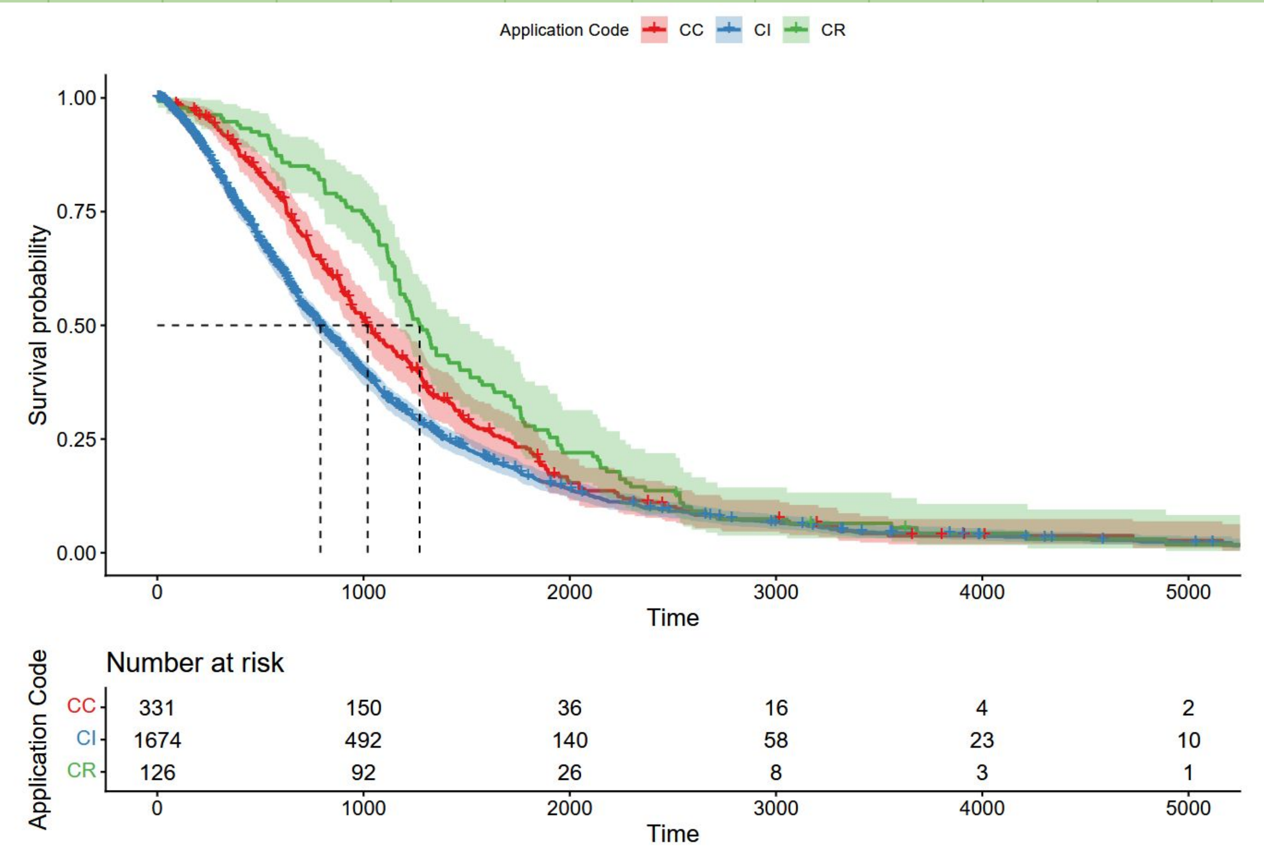
Consumer Loans

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.67417	1.96239	0.12605	5.348	8.88e-08
Monthly_Payment	0.03298	1.03353	0.01672	1.972	0.0486

Survival Curve



Survival Curve After Censoring



Fitting Model with Loan Term

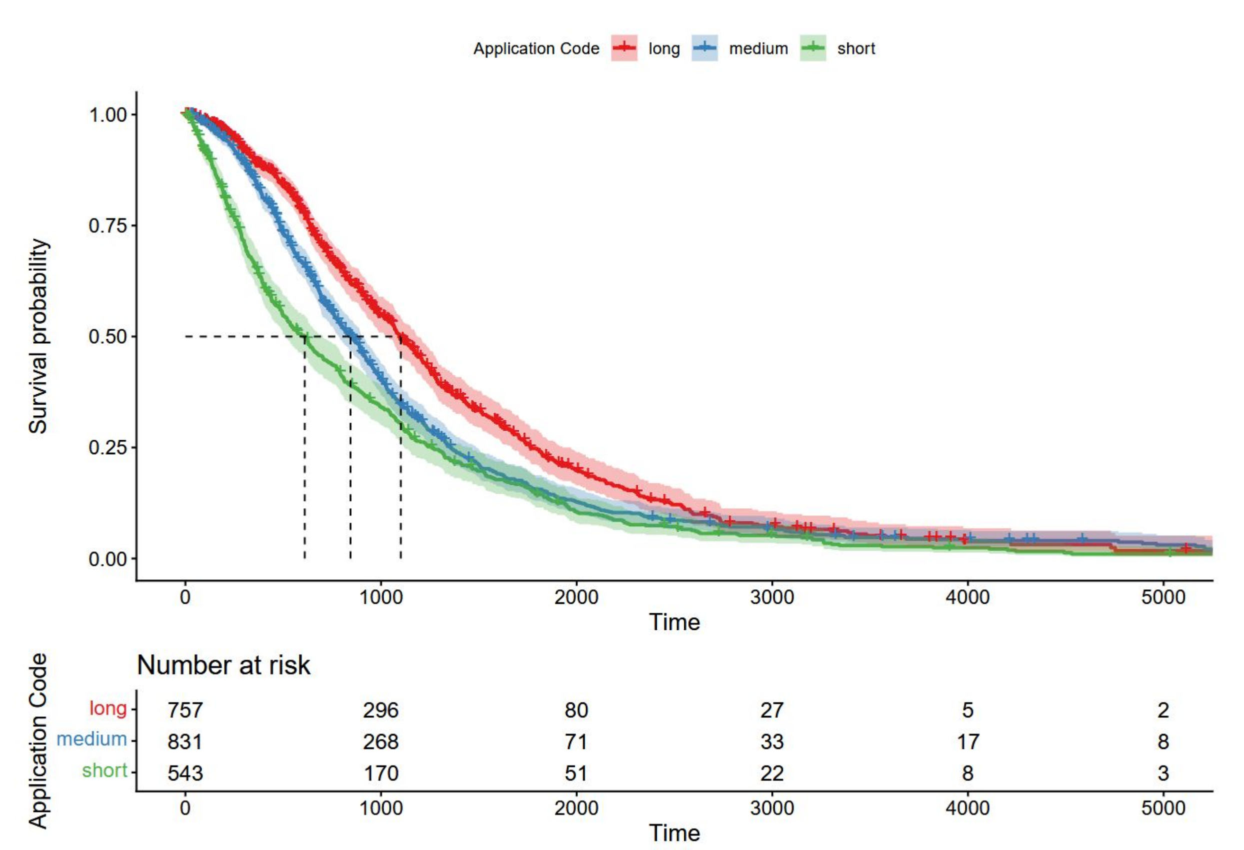
```
Call:
coxph(formula = Surv(Time_to_Charge_Off, Is_Censored) ~ Effective_Annual_Interest_Rate +
    Monthly_Payment + strata(Term), data = logChargeoffClean2)

  n= 2131, number of events= 1685 

                                    coef exp(coef)  se(coef)      z Pr(>|z|)    
Effective_Annual_Interest_Rate  0.453273  1.573453  0.116654  3.886 0.000102 ***
Monthly_Payment                -0.009029  0.991012  0.013680 -0.660 0.509273    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                               exp(coef) exp(-coef) lower .95 upper .95
Effective_Annual_Interest_Rate     1.573     0.6355    1.2519     1.978
Monthly_Payment                    0.991     1.0091    0.9648     1.018
```

Survival Curve

Long: > 5 years

Medium: 2-5 years

Short: < 2 years



Checking the Proportional Hazard Assumption

	chisq	df	p
exp(Effective_Annual_Interest_Rate)	0.605	1	0.44
log(Time_To_Maturity)	81.895	1	<2e-16
GLOBAL	81.935	2	<2e-16

How Does Our Model Perform?

Predicting Hypothetical Loans

Good loan

Interest rate of 2.5%

Monthly payment of $100

```
goodLoan = data.frame( Monthly_Payment = log(100),
    Effective_Annual_Interest_Rate = log(0.025),
    Application_Code = "CI")
```

Bad loan

Interest rate of 25%

Monthly payment of $1000

```
badLoan = data.frame( Monthly_Payment = log(1000),
    Effective_Annual_Interest_Rate = log(0.25),
    Application_Code = "CI")
```

Predicting Hypothetical Loans

The Good Loan: 0.3662471 times less likely to charge off at any given moment than the average Installment or Traditional Consumer Loan.

The Bad Loan: 1.956174 times more likely to charge off at any given moment than the average Installment or Traditional Consumer Loan.

Confidence Interval for the Good Loan

```
                 1
     time      n.risk     n.event    survival     std.err lower 95% CI
  365.000    1197.000     352.000       0.913       0.015        0.884
upper 95% CI
       0.943
```

Interpretation: For an installment or traditional consumer loan with a payment of $100 a month and an effective annual interest rate of 2.5% that will eventually charge off, we are 95% confident that the true probability of the loan not charging off for one year is between 88.4% and 94.3%.

Confidence Interval for the Bad Loan

```
              1
      time        n.risk       n.event      survival       std.err lower 95% CI
   365.000      1197.000       352.000         0.615         0.036        0.549
upper 95% CI
     0.690
```

Interpretation: For an installment or traditional consumer loan with a payment of $1000 a month and an effective annual interest rate of 25% that will eventually charge off, we are 95% confident that the true probability of the loan not charging off for one year is between 54.9% and 69.0%.

Final Model

$$h_g(t; EAR, Monthly\,Payment) = h_{0_g}(t) * e^{0.67445*ln(EAR)+0.05318*ln(Monthly\,Payment)}$$

EAR: Effective annual interest rate

Monthly Payment: the monthly payment that would be made on the loan if it were paid through even payments throughout the life of the loan

$h_g(t)$: hazard at time t for group g

$h_{0_g}(T)$: Baseline hazard for group g, the type of the loan which is either Commercial Loans, Installment or Traditional Consumer Loans, or Real Estate

Final Model Coefficient Interpretation

	coef	exp(coef)	se(coef)	z	p
Effective_Annual_Interest_Rate	0.67313	1.96036	0.11452	5.878	4.16e-09
Monthly_Payment	0.05381	1.05528	0.01476	3.646	0.000267

Likelihood ratio test=38.12 on 2 df, p=5.289e-09
n= 2131, number of events= 1688

EAR: A 1% increase in the EAR is associated with a 0.67345% increase in the hazard of charge-off.

Monthly Payment: A 10% increase in the monthly payment is associated with a 0.5081% increase in the hazard of charge-off.

* We cannot interpret application code because we have stratified it

Conclusion

- Higher interest rate plays a major role in the risk of charge off for loans that will eventually charge off.
- A very large monthly payment will also increase the hazard of charge off.
- Significant separation in application code, so it should also be considered.

Application

- Can be applied to mixed dataset with loans that will and will not be charged off.
- Can be used by loan officers for decision making
- Can use additional variables like credit score, debt-to-income ratio, age, etc.

Any Questions?